

# pyvisco: A Python library for identifying Prony series parameters of linear viscoelastic materials

Martin Springer <sup>1</sup>✉

<sup>1</sup> National Laboratory of the Rockies, United States  ✉ Corresponding author

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## Software

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## Summary

Polymeric materials commonly used in engineering applications, such as the encapsulants and backsheets in photovoltaic modules, adhesives in structural joints, and rubbers in seals and dampers, exhibit time- and temperature-dependent mechanical behavior. The theory of linear viscoelasticity provides a framework to describe this behavior, and the Generalized Maxwell model represented through a Prony series is widely adopted in commercial finite element analysis (FEA) software to simulate viscoelastic response of thermorheologically simple materials under mechanical and thermal loads. Identifying the Prony series parameters from experimental measurements is, however, a non-trivial multi-step process that involves data conditioning, time-temperature superposition (TTSP), shift function fitting, and constrained non-linear least-squares optimization.

pyvisco is an open-source Python library that automates this workflow. Given either raw isothermal measurements at multiple temperatures (from dynamic mechanical thermal analysis or relaxation experiments) or a pre-assembled master curve, the library constructs the master curve via TTSP, fits Williams-Landel-Ferry (WLF) and polynomial shift functions, and identifies the relaxation times  $\tau_i$  and moduli  $E_i$  (or shear moduli  $G_i$ ) of a Prony series compatible with the Generalized Maxwell model. An optional minimization routine produces a reduced-term Prony series suitable for computationally efficient FEA simulations. The library is accessible through a programmatic application programming interface (API) and through a Jupyter-based graphical user interface (LinViscoFit.ipynb) that runs without local installation as a Voila application on Hugging Face Spaces. Library outputs are verified against the curve-fitting routine of ANSYS APDL 2021 R1 (ANSYS Inc., Canonsburg, Pennsylvania) using bundled reference cases.

## Statement of need

Linear viscoelastic material models are routinely required by researchers and engineers working with polymers in structural, thermal, and reliability analyses. The underlying mathematical framework is well established and combines the Boltzmann superposition integral, the Generalized Maxwell model, time-frequency interconversion via Fourier transformation, and TTSP (Brinson & Brinson, 2015; Ferry, 1980; Park & Schapery, 1999; Roylance, 2001; Williams et al., 1955). This framework forms the basis of the characterization workflow described by Springer & Bosco (2020), on which pyvisco is built. However, translating raw experimental data into a Prony series suitable for FEA is tedious and error-prone. The process typically requires:

- shifting isothermal measurement sets in the frequency or time domain to form a master curve at a reference temperature;
- fitting a shift function (e.g., WLF (Williams et al., 1955), Arrhenius, or low-degree polynomial) to parameterize the temperature dependence;

- selecting an appropriate number and distribution of Prony terms;
- solving a bound-constrained least-squares problem to identify  $\tau_i$  and  $E_i$  while enforcing  $E_\infty \geq 0$  and  $\sum \alpha_i \leq 1$ ;
- optionally reducing the number of Prony terms to keep FEA simulations tractable.

Many researchers implement this pipeline ad hoc in spreadsheets or short scripts. Furthermore, commercial dynamic mechanical thermal analysis (DMTA) software (e.g., Netzsch GABO Eplexor, TA Instruments TRIOS) typically allows only a subset of these steps and offers limited interoperability with FEA pre-processors. `pyvisco` was designed to fill this gap with a reproducible, open-source workflow that:

1. accepts both raw and pre-shifted master-curve inputs in CSV or Eplexor Excel format;
2. produces Prony series output cards directly usable in commercial FEA codes;
3. is accessible to non-programmers through a browser-based notebook interface (powered by `ipywidgets`) without local installation; and
4. is verified against ANSYS APDL on a reference benchmark set kept in the repository.

The library targets researchers characterizing polymeric materials (e.g., photovoltaic (PV) encapsulants, adhesives, elastomers, hydrogels) and engineers preparing material cards for viscoelastic FEA simulations.

## State of the field

A number of tools touch parts of the viscoelastic characterization workflow:

- **Commercial DMTA software** (Netzsch GABO Eplexor, TA Instruments TRIOS, Anton Paar RheoCompass) can construct master curves and fit shift functions, but Prony series export and reduced-term optimization for FEA are limited or absent, and licensing restricts reproducibility.
- **pyRheo** (Miranda-Valdez, Coffeng, et al., 2024) focuses on rheological model fitting (fractional and integer-order constitutive models) but does not target Prony-series identification for the Generalized Maxwell model used in FEA, nor does it integrate the full TTSP pipeline.
- **RepTate** (Boudara et al., 2020) is an open-source rheology data analysis suite with a Qt GUI; while powerful for rheological model selection, it is primarily a desktop application and does not export Prony-series cards specifically tuned for finite element use.
- **Bespoke MATLAB/Python scripts** are common in the literature (Barrientos et al., 2019; Park & Schapery, 1999) but are typically not packaged, tested, or maintained beyond a single publication.

`pyvisco` was built as a focused, reusable package because none of the above combine: (a) a full raw-data-to-Prony-series workflow in a single tool, (b) an optimization routine that yields a reduced-term Prony series suitable for FEA, (c) verification against a commercial FEA software's own fitting routine, and (d) a zero-install browser interface that lowers the barrier for non-Python users working in materials labs. Where alternatives are well suited to their target use cases, `pyvisco` complements them by closing the loop between DMTA measurements and FEA input decks.

## Software design

`pyvisco` is organized as a small set of single-purpose modules (`load`, `master`, `shift`, `prony`, `opt`, `out`, `verify`, `styles`, `inter`) that each expose pure functions operating on `pandas.DataFrame` inputs. This functional, dataframe-centric design was chosen so that:

- each step (loading, master-curve assembly, shift-function fitting, Prony fitting, optimization, output) can be invoked, inspected, and unit tested in isolation;

- intermediate states are first-class data artifacts that can be cached, exported, or visualized without re-running upstream computations; and
- the user interface layer (`inter.py`, an `ipywidgets`-based controller) is a thin orchestration layer with no domain logic of its own.

The Prony fit is formulated as a bound-constrained non-linear least-squares problem solved with SciPy's L-BFGS-B implementation (Virtanen et al., 2020), following the formulation of Barrientos et al. (2019) and the interconversion methodology of Park & Schapery (1999). Time-domain to frequency-domain conversion of relaxation data uses the Fourier transformation summarized in Springer & Bosco (2020). Master curves are optionally smoothed using a Savitzky–Golay filter (Savitzky & Golay, 1964) before Prony-series fitting to suppress measurement noise without broadening the relaxation spectrum. The reduced-term optimization routine progressively removes Prony terms and refits, using the ratio of residuals ( $R_{\text{opt}}^2 \approx 1.5 R_0^2$ ) as a default stopping heuristic that the user can override. Numerical routines build on numpy (Harris et al., 2020), scipy (Virtanen et al., 2020), pandas (McKinney, 2010), and matplotlib (Hunter, 2007); the graphical user interface (GUI) is served by Voila as a Docker-based Hugging Face Space.

Outputs include the fitted Prony parameters as CSV, plots in PNG, and a single ZIP archive bundling all results for downstream FEA use. The verification/ directory contains end-to-end notebooks comparing pyvisco fits against ANSYS APDL 2021 R1 reference cases; these notebooks are executed on every push via GitHub Actions to guard against regressions.

## Research impact statement

pyvisco originated to support viscoelastic characterization of PV module encapsulants and interconnect adhesives, as part of the reliability and durability research described by Springer & Bosco (2020). Since its initial public release in 2022 the package has been published on PyPI, archived on Zenodo with a DOI per release (Springer, 2022), indexed by Read the Docs, and developed openly on GitHub with continuous integration, an issue tracker, a tagged release history, and a public changelog spanning the v1.x and v2.x series.

The v2.0 release re-organized the codebase into the modular layout described above, added a verified test suite, modernized the packaging (PEP 621 `pyproject.toml`, Python 3.11), and introduced standalone tutorial scripts in addition to the interactive notebook so the library can be used in headless and continuous-integration contexts.

Beyond its originating use case, pyvisco has been adopted by independent research groups across a range of polymer-mechanics domains. To date the software has been used in at least eight peer-reviewed publications and technical reports, including: fractional-calculus modeling of methylcellulose aqueous systems (Miranda-Valdez, Puente-Córdova, et al., 2024); finite-strain viscoelastic-damage modeling of solid propellants (Gouhier et al., 2024); characterization and modeling of waterborne epoxy varnish coatings for steel/epoxy laminates in electric machines (Tiefenthaler et al., 2025); structural-dynamic characterization of polymeric foams via time-temperature superposition (Le Barbenchon et al., 2025); fatigue testing of bitumen binders using column specimens (Mangalath Shine et al., 2023); nano-indentation DMTA characterization of polymeric materials in microelectronic packaging (Lin et al., 2024); viscoelastic characterization of polymer films (Lin & Gamez, 2025); and biomechanical modeling of handball-induced head injuries (Johansson & Nilsson, 2025). These applications span food science, aerospace propellants, electric-machine manufacturing, cellular polymers, road-pavement engineering, microelectronics packaging, and sports biomechanics, demonstrating that the workflow generalizes well beyond its original PV context.

By providing a reproducible, open, and FEA-oriented Prony identification workflow, pyvisco complements existing rheology packages and lowers the barrier for materials researchers to share viscoelastic material cards in a form directly usable by the wider engineering community.

## AI usage disclosure

Generative AI tools were used during the preparation of the v2.x release and of this manuscript:

- **Tools/models used:** GitHub Copilot in Visual Studio Code (Claude Opus 4.7 and GPT-class models accessed through Copilot Chat), used during development between 2024 and 2026.
- **Where used:** (a) refactoring of the v1.x codebase into the modular layout, including renaming and reorganization of functions; (b) drafting and updating user-facing documentation (README.md, the changelog under docs/changelog/, docstrings, tutorial scripts); (c) drafting unit-test scaffolding under tests/; (d) initial drafting of this paper.md.
- **Where not used:** the core numerical formulation of the Prony fit, shift-function fits, and time-temperature superposition algorithms predate the use of AI assistance and were authored by the human author; the verification notebooks against ANSYS APDL and their reference data were authored manually.
- **Verification:** All AI-assisted code was reviewed by the author before commit, executed locally, and validated by the existing test suite and the end-to-end verification notebooks comparing pyvisco outputs against ANSYS APDL. AI-assisted prose was reviewed and edited by the author for accuracy. The author takes full responsibility for the correctness, originality, and licensing of all materials in this submission.

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